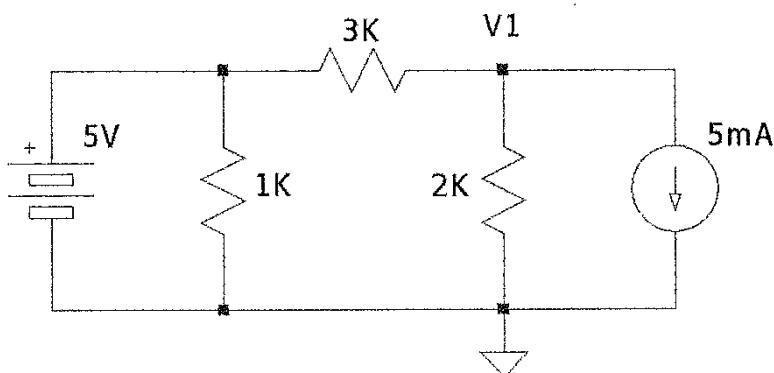


Question 1 [25 Points]:



Find V_1 using **two** of the methods listed below.

Node analysis/ Superposition / Thevenin equivalent / Norton equivalent

(Hint: Check that the two methods give you identical results.)

(a) [12.5 points] Find V_1 using the first method of your choice.

Node Analysis: $\frac{V_1 - 5}{3K} + \frac{V_1}{2K} + 5mA = 0$

$$V_1 \left(\frac{1}{3} + \frac{1}{2} \right) = \frac{5}{3} - 5 = -\frac{10}{3}$$

$$\Rightarrow \boxed{V_1 = -4V}$$

superposition:

with current source killed:

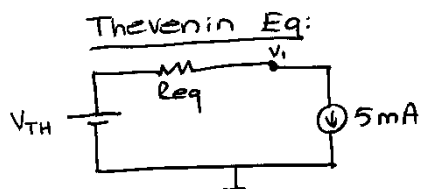
$$V_{1,1} = 5 \cdot \frac{2}{2+3} = 2V$$

with voltage source killed:

$$V_{1,2} = -5mA \cdot (2K // 3K) = -5 \cdot \frac{6}{5} = -6V$$

$$V_1 = V_{1,1} + V_{1,2} = 2V - 6V = \boxed{-4V}$$

(b) [12.5 points] Find V_1 using the second method of your choice.

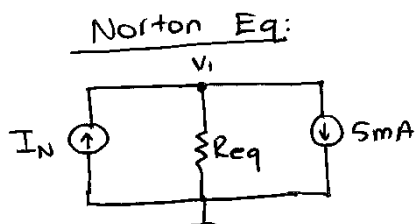


$$V_{TH} = 5 \cdot \frac{2K}{2K+3K} = 2V$$

$$R_{eq} = 3K // 2K = \frac{6}{5} K$$

$$\left. \begin{array}{l} \text{Then,} \\ V_1 = V_{TH} - 5mA \cdot R_{eq} \\ = 2 - 5 \cdot \frac{6}{5} \end{array} \right\}$$

$$\boxed{V_1 = -4V}$$



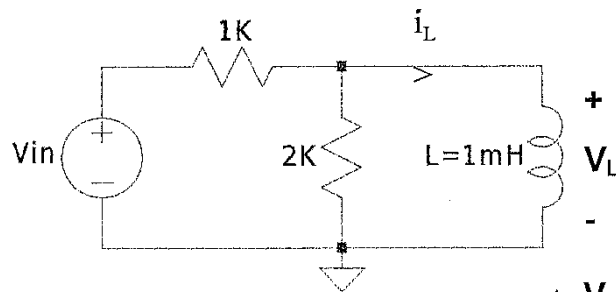
$$I_N = \frac{5V}{3K} = \frac{5}{3} mA$$

$$R_{eq} = 3K // 2K = \frac{6}{5} K$$

$$\left. \begin{array}{l} \text{Then,} \\ V_1 = (I_N - 5mA) \cdot R_{eq} \\ = \left(\frac{5}{3} - 5 \right) \cdot \frac{6}{5} \end{array} \right\}$$

$$\boxed{V_1 = -4V}$$

Question 2 [25 points]:



For the circuit shown above, the voltage source $V_{in}(t)$ is as shown on the right.

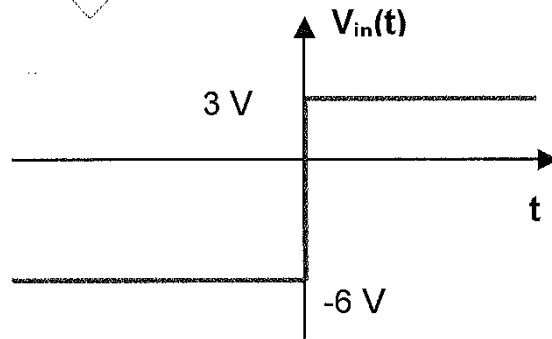
(a) [5 points] Find $i_L(0^-)$ and $V_L(0^-)$.

(0^- denotes the time right before $t = 0$).

L is assumed short circuit.

$$V_L(0^-) = 0 \text{ V}$$

$$i_L(0^-) = \frac{-6 \text{ V}}{1 \text{ k}} = -6 \text{ mA}$$



(b) [10 points] Find $i_L(0^+)$ and $V_L(0^+)$.

(0^+ denotes the time right after $t = 0$).

inductor cannot change its current instantaneously.

$$i_L(0^+) = -6 \text{ mA}$$

Replace inductor with a current source



$$\frac{V_L - 3}{1 \text{ k}} + \frac{V_L}{2 \text{ k}} = 6 \text{ mA}$$

$$3V_L - 6 = 12$$

$$V_L(0^+) = 6 \text{ V}$$

* Eventually, inductor will act like a short circuit

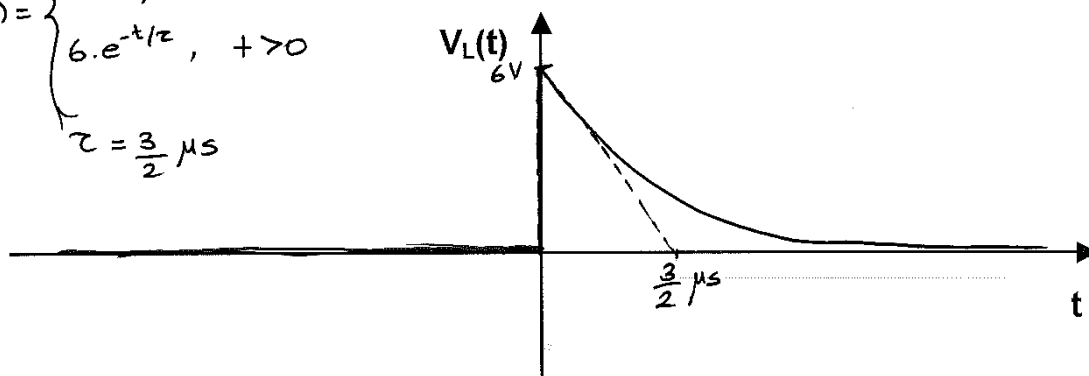
$$V_L(\infty) = 0 \text{ V}$$

$$\tau = \frac{L}{R} = \frac{1 \text{ mH}}{(1 \text{ k} \parallel 2 \text{ k})} = \frac{1 \text{ mH}}{2/3 \text{ k}} = \frac{3}{2} \mu \text{ s} \quad \left. \begin{array}{l} \text{for } t > 0, \\ V_L(t) = 6 \cdot e^{-t/\tau} \end{array} \right\}$$

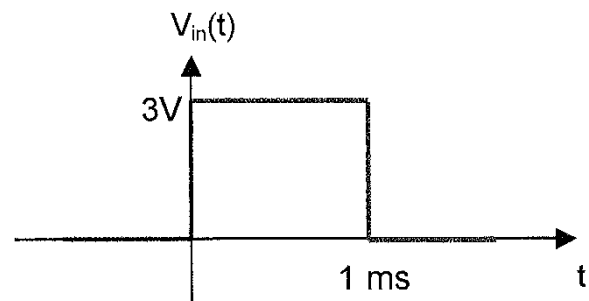
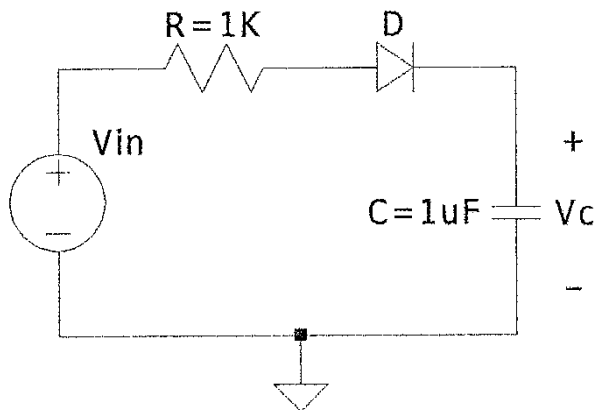
(c) [10 points] Find and plot $V_L(t)$ for $-\infty < t < \infty$. Plot it onto the axes provided below. Mark the scales of the plot and indicate all units properly.

$$V_L(t) = \begin{cases} 0, & t < 0 \\ 6 \cdot e^{-t/\tau}, & t > 0 \end{cases}$$

$$\tau = \frac{3}{2} \mu \text{ s}$$



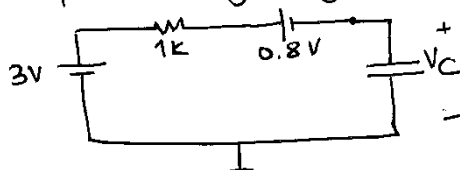
Question 3 [25 Points]:



In the circuit above, $C = 1\mu\text{F}$, $V_C(0) = 0\text{ V}$, and the threshold voltage of the diode is $V_{ON} = 0.8\text{ V}$. The input voltage $V_{in}(t)$ is plotted above.

Find $V_C(t)$ for $t > 0$. Plot it onto the axes provided below. Mark the scales of the plot and indicate all units properly. Specify the value of $V_C(t)$ at $t = 1\text{ ms}$.

Input voltage greater than V_{ON} . So, D is ON.



$$V_C(\infty) = 3\text{ V} - 0.8\text{ V} = 2.2\text{ V}$$

$$V_C(0) = 0\text{ V}$$

$$\tau = RC = 1\mu\text{F} \cdot 1\text{K}\Omega = 1\text{ ms}$$

$$V_C(t) = 2.2 + (0 - 2.2)e^{-t/\tau} = 2.2(1 - e^{-t/\tau})$$

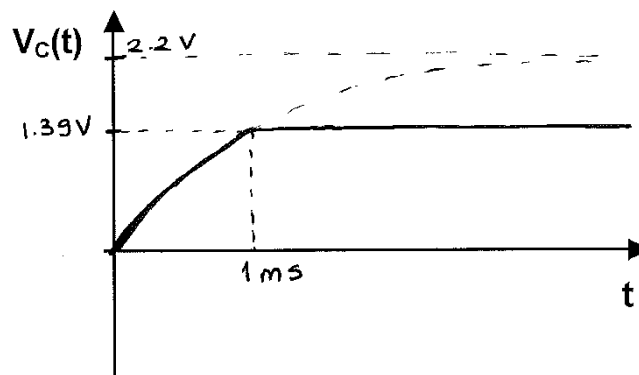
* But at $t = 1\text{ ms}$, V_{in} drops to zero

$\Rightarrow$ D turns OFF

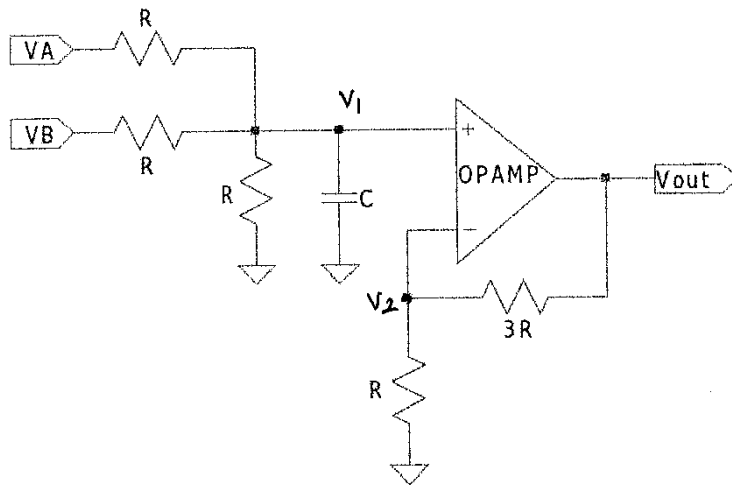
C cannot discharge, so it will keep its voltage

$$V_C(t = 1\text{ ms}) = 2.2(1 - e^{-1/1}) = 1.39\text{ V}$$

$$\text{So, } V_C(t) = \begin{cases} 2.2(1 - e^{-t/\tau}) & , 0 < t < 1\text{ ms} \\ 1.39\text{ V} & , t \geq 1\text{ ms} \end{cases}, \tau = 1\text{ ms}$$



Question 4 [25 Points]:



For the circuit shown above, assume that V_{out} is not saturated.

(a) [10 points] With C removed from the circuit ($C = 0$), find V_{out} in terms of V_A and V_B .

$$V_1 = V_2$$

$$V_1 = V_A \cdot \frac{R/2}{R/2 + R} + V_B \cdot \frac{R/2}{R/2 + R} = (V_A + V_B) \cdot \frac{1}{3}$$

$$V_2 = V_{out} \cdot \frac{R}{R + 3R} = V_{out} \cdot \frac{1}{4}$$

$$\frac{1}{4} V_{out} = (V_A + V_B) \cdot \frac{1}{3} \Rightarrow \boxed{V_{out} = \frac{4}{3} (V_A + V_B)}$$

(b) [10 points] With $V_B = 0$ and $C \neq 0$ (C is not removed), find the transfer function

$$H(\omega) = \frac{V_{out}}{V_A}, \text{ assuming sinusoidal input signal.}$$

$$V_1 = \frac{(R/2) \parallel (\frac{1}{j\omega C})}{(R/2) \parallel (\frac{1}{j\omega C}) + R} \cdot V_A = V_A \cdot \frac{\frac{R}{2 + j\omega RC}}{\frac{R}{2 + j\omega RC} + R} = V_A \cdot \frac{R}{R + 2R + j\omega R^2 C}$$

$$V_1 = V_A \cdot \frac{1}{3 + j\omega RC}$$

$$V_2 = V_{out} \cdot \frac{R}{R + 3R} = \frac{V_{out}}{4}$$

$$V_1 = V_2 \Rightarrow V_A \cdot \frac{1}{3 + j\omega RC} = \frac{V_{out}}{4}$$

$$H(\omega) = \frac{V_{out}}{V_A} = \frac{4}{3 + j\omega RC} = \boxed{\frac{4}{3} \cdot \frac{1}{1 + j\omega \frac{R}{3} C}}$$

(c) [5 points] Find the 3-dB corner frequency, f_c , of the transfer function from part (b).

$$\text{At 3-dB frequency, } |H(\omega)| = \frac{4}{3\sqrt{2}} \Rightarrow 1 = j\omega \frac{R}{3} C$$

$$\omega_c = \frac{3}{RC}$$

$$\boxed{f_c = \frac{3}{2\pi RC}}$$